

LOWIN CAPSULE 300 MG

DESCRIPTION:

LOWIN CAPSULE 300MG: White / red capsule with marking 'DUO 861'.

COMPOSITION:

LOWIN CAPSULE 300MG: Each capsule contains Gemfibrozil 300 mg.

PHARMACODYNAMICS:

Gemfibrozil is a lipid regulating agent which lowers elevated serum lipid primarily by decreasing serum triglyceride with a variable reduction in total serum cholesterol. These decreases occur primarily in the very low density lipoprotein (VLDL) fraction and less frequently in the low density lipoprotein (LDL) fraction. It has been reported that Gemfibrozil may increase levels of high density lipoprotein (HDL) sub fraction, HDL₂ and HDL₃ as well as apolipoproteins A1 and A11. Epidemiological studies have shown that both HDL-cholesterol and high LDL-cholesterol are independent risk factors for coronary heart disease. The mechanism of action has not been definitively established. In man, Gemfibrozil has been shown to inhibit peripheral lipolysis and to decrease the hepatic extraction of free fatty acids, thus reducing hepatic triglyceride production. Gemfibrozil inhibits synthesis and increases clearance of VLDL carrier apolipoprotein B, leading to a decrease in VLDL production. Animal studies suggest that Gemfibrozil may, in addition to elevating HDL cholesterol, reduce incorporation of long-chain fatty acids into newly formed triglycerides, accelerate turnover and removal of cholesterol from the liver and increase excretion of cholesterol in the feces.

PHARMACOKINETICS:

Gemfibrozil is well absorbed from the gastrointestinal tract after oral administration. Peak plasma levels occur in one to two hours with a plasma half-life of 1.5 hours following single doses and 1.3 hours following multiple doses. Plasma levels appear proportional to dose and do not demonstrate accumulation across time following multiple doses. Gemfibrozil mainly undergoes oxidation of a ring methyl group to successively form a hydroxymethyl and carboxyl metabolite. Approximately seventy percent of the administered human dose is excreted in the urine, mostly as the glucuronide conjugate, with less than 2% excreted as unchanged Gemfibrozil. Six percent of the dose is accounted for in the feces.

INDICATION:

The initial treatment for hyperlipidemia is dietary therapy specific for the type of hyperlipoproteinemia. Drug therapy should not be used for the routine treatment of elevated blood lipids for the prevention of coronary heart disease. Excess body weight and excess alcoholic intake may be important factors in hypertriglyceridemia and should be addressed prior to any drug therapy. Physical exercise can be an important ancillary measure. Contributory diseases such as hypothyroidism or diabetes mellitus should be looked for and adequately treated. The use of drugs should be considered only when reasonable attempts have been made to obtain satisfactory results with nondrug methods. If the decision is to use drugs, the patient should be instructed that this does not reduce the importance of adhering to diet. Gemfibrozil may be considered for the treatment of adult patients with very high serum triglyceride levels (type IV or type V hyperlipoproteinemia) who present a risk of abdominal pain and pancreatitis and who do not respond adequately to a determined dietary effort to control them. Patients with triglyceride levels in excess of 750 mg per deciliter are likely to present such risk. Gemfibrozil has little effect on elevated cholesterol levels in most subjects. A minority of subjects show a more pronounced response. Therefore, the physician should be very selective and confine gemfibrozil treatment to patients with clearly defined risk due to severe hypercholesterolemia (eg. individuals with familial hypercholesterolemia starting in childhood) who inadequately respond to appropriate diet and more effective cholesterol drugs. Gemfibrozil is not useful for the hypertriglyceridemia of Type I hyperlipidemia. The biochemical response to gemfibrozil is variable, and it is not always possible to predict from the lipoprotein type or other factors which patients will obtain favourable results. It is essential that lipid levels be assessed and that the drug be discontinued after three months in any patient in whom lipid do not show significant improvement. The effect of drug-induced reduction of serum cholesterol or triglyceride levels or elevation of HDL cholesterol levels on morbidity or mortality due to coronary heart disease has not been established. Several years may be required before ongoing long-term investigations will resolve this question.

RECOMMENDED DOSAGE:

The recommended dose for adults is 1200 mg administered in two divided doses 30 minutes before the morning and evening meal. Some patients will experience therapeutic effects on 900 mg/day; a few require 1500 mg/day for satisfactory results.

ROUTE OF ADMINISTRATION: For oral use.

CONTRAINDICATIONS:

1. Hepatic or severe renal dysfunction, including primary biliary cirrhosis.
2. History of/ or pre-existing gall bladder or biliary tract disease, including gallstones.
3. Hypersensitivity to gemfibrozil.
4. Concomitant use of repaglinide or simvastatin.
5. Patients with previous history of photoallergy or phototoxic reaction during treatment with fibrates.

WARNING AND PRECAUTIONS:

Muscle disorders (myopathy/rhabdomyolysis):

There have been reports of myositis, myopathy and markedly elevated creatine phosphokinase associated with gemfibrozil. Rhabdomyolysis has also been reported rarely.

Muscle damage must be considered in any patient presenting with diffuse myalgia, muscle tenderness and/or marked increase in muscle CPK levels (>5x ULN); under these conditions treatment must be discontinued.

Use in patients with gallstone formation:

Gemfibrozil may increase cholesterol excretion into the bile raising the potential for gallstone formation. Cases of cholelithiasis have been reported with gemfibrozil therapy. If cholelithiasis is suspected, gallbladder studies are indicated. Gemfibrozil therapy should be discontinued if gallstones are found.

Monitoring serum lipids:

Periodic determinations of serum lipids are necessary during treatment with gemfibrozil. Sometimes a paradoxical increase of (total and LDL) cholesterol can occur in patients with hypertriglyceridaemia. If the response is insufficient after 3 months of therapy at recommended doses treatment should be discontinued and alternative treatment methods considered.

Monitoring liver function:

Elevated levels of ALAT, ASAT, alkaline phosphatase, LDH, CK and bilirubin have been reported. These are usually reversible when gemfibrozil is discontinued. Therefore liver function tests should be performed periodically. Gemfibrozil therapy should be terminated if abnormalities persist.

Monitoring blood counts:

Periodic blood count determinations are recommended during the first 12 months of gemfibrozil administration. Anaemia, leucopenia, thrombocytopenia, eosinophilia and bone marrow hypoplasia have been reported rarely.

Use in children: Safety and efficacy in children have not been established.

INTERACTIONS WITH OTHERS MEDICAMENTS:

Concomitant use with CYP2C8, CYP2C9, CYP2C19, CYP1A2, UGT1A1 and UGT3A3 substrates:

The interaction profile of gemfibrozil is complex resulting in increased exposure of many medicinal products if administered concomitantly with gemfibrozil.

Gemfibrozil potently inhibits CYP2C8, CYP2C9, CYP2C19, CYP1A2, UGT1A1 and UGT3A3 enzymes.

Concomitant use with hypoglycaemic agents:

There have been reports of hypoglycaemic reactions after concomitant use with gemfibrozil and hypoglycaemic agents (oral agents and insulin). Monitoring of glucose levels is recommended.

Concomitant anticoagulants:

Gemfibrozil may potentiate the effects of coumarin type vitamin K antagonist anticoagulants such as warfarin, acenocoumarol, or phenprocoumon. The concomitant administration of gemfibrozil with these anticoagulants necessitates careful monitoring of prothrombin time (INR - International Normalised Ratio). Caution should be exercised when such a coumarin type vitamin K antagonist anticoagulant is given concomitantly with gemfibrozil. The dosage of the anticoagulant may need to be reduced to maintain desired prothrombin time level.

Repaglinide:

The combination of gemfibrozil with repaglinide is contra-indicated.

Rosiglitazone:

The combination of gemfibrozil with rosiglitazone should be approached with caution.

HMG CoA reductase inhibitors:

The concomitant administration of gemfibrozil with simvastatin is contraindicated. The combined use of gemfibrozil and a statin should generally be avoided. The use of fibrates alone is occasionally associated with myopathy.

There have been reports of severe myositis with markedly elevated creatine kinase and myoglobinuria (rhabdomyolysis) when gemfibrozil and HMG CoA reductase inhibitors were used concomitantly and dosage adjustments may be necessary.

The benefit of further alterations in lipid levels by the combined use of gemfibrozil and HMG-CoA reductase inhibitors should be carefully weighed against the potential risks of such combinations and clinical monitoring is recommended.

Bexarotene:

Concomitant administration of gemfibrozil with bexarotene is not recommended.

Bile Acid – Binding Resins:

Reduced bioavailability of gemfibrozil may result when given simultaneously with resin-granule drugs such as colestipol. Administration of the products two hours or more apart is recommended.

Colchicine:

Risk of myopathy and rhabdomyolysis may be increased with concomitant administration of colchicine and gemfibrozil. This risk may be increased in the elderly and in patients with hepatic or renal dysfunction.

Gemfibrozil is highly bound to plasma proteins and there is potential for displacement interactions with other drugs.

USE IN PREGNANCY & LACTATION:

Pregnancy category B: Reproduction studies have been performed in the rat at doses 3 and 9 times the human dose, and in the rabbit at 2 and 6.7 times the human dose. These studies have revealed no evidence of impaired fertility in females or harm to the fetus due to Gemfibrozil. Minor fetotoxicity was manifested by reduced birth rates observed at the high dose levels. No significant malformations were found among almost 400 offspring from 36 litters of rats and 100 fetuses from 22 litters of rabbits. There are no studies in pregnant women. In view of the fact that Gemfibrozil is tumorigenic in male rats, the use of Gemfibrozil in pregnancy should be reserved for those patients where the benefit clearly outweighs the possible risk to the patient or fetus.

Nursing mothers: Because of the potential for tumorigenicity shown for gemfibrozil in male rats, a decision should be made whether to discontinue nursing or discontinue the drug, taking into account the importance of the drug to the mother.

SIDE EFFECTS/ ADVERSE EFFECTS:

Most commonly reported adverse reactions are of gastrointestinal character and are seen in approximately 7% of the patients. These adverse reactions do not usually lead to discontinuation of the treatment.

Skin and subcutaneous tissue disorders: rash, dermatitis, pruritus, urticarial, eczema

Central Nervous System: headache, dizziness, blurred vision, vertigo, insomnia, paresthesia, tinnitus

Musculoskeletal: painful extremities

Hematopoietic: anaemia, eosinophilia, leukopenia.

Gastrointestinal: dry mouth, abdominal pain constipation, anorexia, gas pain, dyspepsia, diarrhea, vomiting, nausea, flatulence

Musculoskeletal: backpain, arthralgia, muscle cramps, myalgia, swollen joints, myositis

Miscellaneous: fatigue, malaise, syncope, alopecia.

SYMPTOMS AND TREATMENT OF OVERDOSE:

While there has been no reported case of overdosage, symptomatic supportive measures should be taken should it occur.

STORAGE CONDITIONS:

Store in a cool dry place below 30°C.

Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZE:

Blister pack of 3 x 10's capsules in a box.

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599 Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang,

Selangor, MALAYSIA.